Elita Lestari Setianto (201344008)	**Rangkuman Minggu 3**	Senin, 12 September 2022
		16TNK5043 (Aplikasi Mikrokontroler)
3NK		Ferry Satria, BSEE.,M.T. Rahmawati Hasanah,S.ST.,M.T.

LAB MINGGU KE 3

1. **Konversi Data, Desimal ke Oktal**

Menggunakan bagi 8

Contoh

1. 65(10) = 8 sisa 1
   8 /8 = 1 sisa 0
   65 = 101

2. 92(10) = 11 sisa 4
   11 / 8 = 1 sisa 3
   92 = 134

Cara menentukan algoritma pemograman

- menggunakan pengurangan bilangan (menggunakan sub rutin) Bilangan dikali 0A
- Pengurangan decimal menggunakan complement 10
- Di ubah ke heksadesimal kemudian di disisipkan 0 di 2 sisi

2. **Konversi Data, Oktal ke Desimal**

Cara menentukan algoritma pemograman

- Menggunakan sub rutin perkalian 8
- Menggunakan satuan, delapanan, enampuluh empatan

Kode Program

```
.include "m8535def.inc"
.org 0x00
rjmp main
        ; PAKAI PROJECT "ROOT"
```

```
main :
ldi r20,low(ramend)
out spl,r20
ldi r20,high(ramend)
out sph,r20

ldi r20,0x44            ; input data dari data area
mov r21,r20
andi r21,0xf0
lsr r21
lsr r21
lsr r21
lsr r21
rcall gnd
mov r22,r21
mov r21,r20
andi r21,0x0f
add r21,r22
mov r19,r21
rcall daa
mov r21,r19



akhir: rjmp akhir

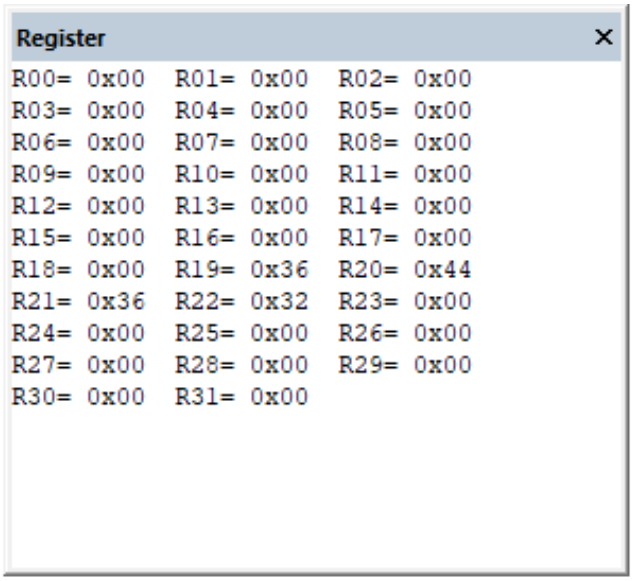
gnd:
ldi r16,0x03
l1:     add r21,r21
        mov r19,r21
        rcall daa
        mov r21,r19
        dec r16
```

```
        brne l1
ret

; subrutin unt koreksi hasil penjumlahan mjd. BCD.2 dgt.
; yang dilaksanakan mll. r19 sbg. acc.:
daa:    push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
```

```
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:    out sreg,r16
                adc r24,r24
                ret
fk_hn:  ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```

Hasil Program



- **Program (Metode Lookup Table)**

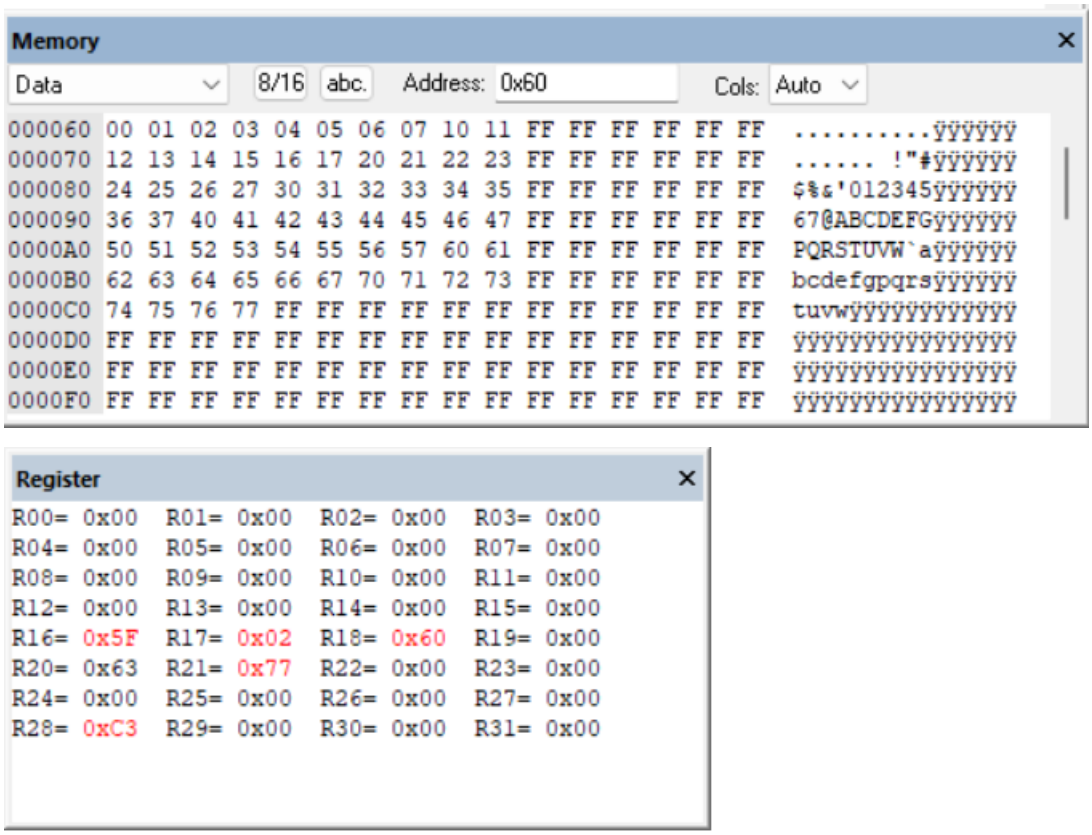
```
.include "m8535def.inc"
.org 0x00
rjmp main

main :
ldi r16,low(ramend)
out spl,r16
ldi r17,high(ramend)
out sph,r17

ldi r18,0X60
mov r28,r20
add r28,r18
adc r29,r30
ld r21,y

akhir : rjmp akhir
```

- **Hasil**



1. Buat program Konversi Data Oktal 2 digit (yang ada di R20) ke Desimal. Simpan hasil konversi pada R21.

- **Program**

```
.include "m8535def.inc"
.org 0x00
rjmp main
    ; PAKAI PROJECT "ROOT"
main :
ldi r20,low(ramend)
out spl,r20
ldi r20,high(ramend)
out sph,r20
```

```
ldi r20,0x44                ; data input
mov r22,r20
andi r22,0xf0               ; mengambil nilai delapanan
lsr r22
lsr r22
lsr r22
lsr r22
rcall kal                   ; penggandaan 3 kali
mov r21,r22

mov r22,r20
andi r22,0x0f

add r21,r22
mov r19,r21
rcall daa
mov r21,r19

akhir: rjmp akhir

kal:        ; subrutin perkalian 8 secara desimal = penggandaan secara 3
kali
ldi r16,0x03
l1:  add r22,r22
     mov r19,r22
     rcall daa
     mov r22,r19
     dec r16
     brne l1
ret

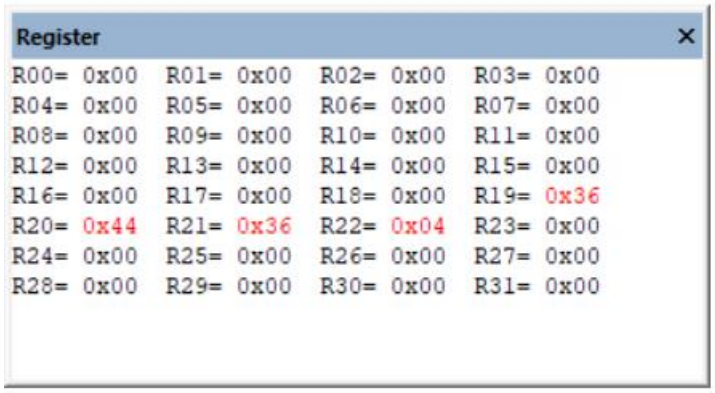
daa: push r16;amankan isi reg. ke stack
```

```
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi


; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:           mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:           out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:           mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:           out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
```

```
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:          ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh: out sreg,r16
                adc r24,r24
                ret
fk_hn:          ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```

- **Hasil**


Register
R00= 0x00 R01= 0x00 R02= 0x00 R03= 0x00
R04= 0x00 R05= 0x00 R06= 0x00 R07= 0x00
R08= 0x00 R09= 0x00 R10= 0x00 R11= 0x00
R12= 0x00 R13= 0x00 R14= 0x00 R15= 0x00
R16= 0x00 R17= 0x00 R18= 0x00 R19= 0x36
R20= 0x44 R21= 0x36 R22= 0x04 R23= 0x00
R24= 0x00 R25= 0x00 R26= 0x00 R27= 0x00
R28= 0x00 R29= 0x00 R30= 0x00 R31= 0x00